

Ismail Elyan

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer and AWS-Certified cloud architect with hands-on expertise in Large Language Models (LLMs), LLMOps, Retrieval-Augmented Generation (RAG), and agentic AI workflows. Proficient in designing scalable, production-ready AI systems on AWS (SageMaker, Bedrock, Lambda, ECS) and building intelligent automation pipelines using n8n integrated with AI agents. Experienced with LangChain, LangGraph, CrewAI, Hugging Face, and PyTorch for end-to-end AI development. Strong background in MLOps, CI/CD for ML, vector databases, and multi-agent orchestration.

WORK EXPERIENCE

ML Trainee – Digital Egypt Pioneers Initiative (DEPI)

Oct 2024 – Jun 2025

- Completed a 9-month intensive AI and Machine Learning program covering NLP, Computer Vision, and end-to-end ML pipelines.
- Built, trained, and evaluated 10+ NLP and Computer Vision models using Python and PyTorch, achieving up to 91% accuracy on benchmark datasets.
- Designed full ML pipelines including data preprocessing, feature engineering, model evaluation, and deployment.
- Deployed models on Azure and Hugging Face Spaces, reducing inference latency by 25% through optimization techniques.

NLP Trainee – National Telecommunication Institute (NTI)

Aug 2025

- Completed 120 hours of intensive NLP training, achieving a 98% final score — top performer in the cohort.
- Developed RAG-based NLP applications using embedding models and vector databases for document retrieval and QA across 5+ use cases.
- Implemented text preprocessing, chunking, and similarity search pipelines, improving response relevance by 35% over baseline.

Cloud Services Trainee – National Telecommunication Institute (NTI)

Oct 2025 – Dec 2025

- Gained hands-on experience with 8 core AWS services including EC2, S3, IAM, VPC, Lambda, RDS, and CloudWatch.
- Designed secure, scalable cloud architectures aligned with AWS Solutions Architect best practices.
- Developed cost-optimized IAM policies and networking configurations, reducing projected cloud costs by 20%.

EDUCATION

B.Sc. in Computer Science – Major in AI and Machine Learning

2022 – 2026

Benha National University | Graduating GPA: 3.65 / 4.0

PROJECTS

AI-Powered Academic Advisor Chatbot

- Engineered a production-grade AI academic advisor chatbot for Benha National University, deployed as a REST API (FastAPI + Uvicorn) integrated into a university mobile app, serving students across 3 program tracks (AI & ML, Software Development, Data Science).
- Built a stateful LangGraph ReAct agent with a custom Judging Loop, orchestrating 12 LangChain tools across Neo4j knowledge graph (course catalog, prerequisites, program tracks), Supabase (student profiles, chat history), and a Pinecone RAG pipeline for university bylaw Q&A — achieving accurate multi-hop reasoning over the full academic domain.

- Designed a 5-step NLP preprocessing pipeline (reference resolution, entity extraction, fuzzy course name mapping, track normalization, query rewriting) that resolves abbreviations and pronouns before the agent runs, eliminating wasted tool calls and handling ambiguous queries via interactive disambiguation sessions.
- Implemented a RAG pipeline using HuggingFace multilingual-e5-large embeddings and Pinecone vector store (bnu-bylaws index) for bylaw/regulation Q&A, and built an interactive multi-turn course planner bridged from a CLI function to the chatbot via a thread I/O bridge with 90-second timeout handling.
- **Stack:** LangGraph, LangChain, FastAPI, Groq, Neo4j Aura, Supabase (PostgreSQL), Pinecone, HuggingFace, Python, Docker.

Nestora – AI Floor Plan Furnishing App (Flutter + FastAPI)

- Built a full-stack Flutter application (Provider state management, Supabase auth/database/storage) that takes a room floor plan photo through a multi-step design wizard — dimensions and door/window placement, style, and furniture preferences — and returns an AI-furnished 2D CAD-style layout.
- Engineered the FastAPI AI backend (nestora_ai/) end to end: a Groq vision model (Llama-3.2-11B-Vision) detects and OCRs room labels on the uploaded plan, OpenCV flood-fill segments room polygons from a single seed point, and a Groq LLM (Llama-3.3-70B) plans furniture placement that is rendered as color-coded CAD symbols via OpenCV/PIL.
- Exposed the pipeline as 4 REST endpoints (/analyze-floorplan, /detect-room, /furnish-floorplan, /chat) tunneled through ngrok for mobile integration, plus a standalone HTML/JS test harness for iterating on the backend independently of the app.
- Wired a Supabase-backed workshop-request flow so a generated design and its spec sheet can be sent directly to a furniture workshop for fabrication, closing the loop from AI design to real-world order.
- **Stack:** Flutter, Provider, Supabase (Auth/Postgres/Storage), FastAPI, Groq API, OpenCV, PIL, Python, ngrok.

BNU Admission Chatbot – Bilingual RAG Assistant for the University Admission Website

- Built a bilingual (Arabic/English) RAG chatbot answering prospective-student admission questions for Benha National University, served through a FastAPI backend with a lightweight web frontend.
- Implemented hybrid retrieval combining a Pinecone dense vector index (bge-m3 1024-dim embeddings via the Hugging Face Inference API) with local BM25 sparse search fused via Reciprocal Rank Fusion, plus an optional Cohere reranker, for accurate answers over admissions policy documents.
- Built an automatic Google Drive ingestion pipeline (service-account auth) that polls a shared folder every 10 minutes, chunks new or modified PDFs, upserts them into Pinecone, and prunes vectors for files deleted from Drive — tracked via a local sync registry keyed on file ID and checksum.
- Served answers with Groq LLMs (gpt-oss-120b, with Llama-3.3-70B fallback) and included a ChromaDB fallback vector store for fully offline development.
- **Stack:** FastAPI, Pinecone, Hugging Face Inference API, Groq, BM25, Cohere, Google Drive API, Python, Docker.

CERTIFICATIONS

- AWS Certified AI Practitioner | AWS Certified Machine Learning Associate
- AWS Certified Cloud Practitioner | AWS Certified Solutions Architect Associate

SKILLS

LLMs & Agentic AI: Hugging Face Transformers, LangChain, LangGraph, CrewAI, RAG, n8n, Prompt Engineering, Fine-tuning, LangSmith, Weights & Biases, Ollama, vLLM

Programming: Python (Advanced), C++ (Intermediate), C# (Beginner)

Orchestration & Deployment: FastAPI, Streamlit, Docker, Kubernetes (EKS), GitHub Actions

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn

Databases: MySQL, Neo4j, MongoDB, Pinecone, Elasticsearch, Redis

LANGUAGES

Arabic – Native | English – Advanced